HyperVerge – Problem 1
Lightweight KYC App for Bharat

Product &
Technical Design

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Context & Target Users

- Audience: rural & semi-urban first-time smartphone users, low literacy, low-end devices, patchy networks
- Goals: frictionless onboarding, trust & safety, regulatory compliance (Digilocker + doc KYC + face auth)
- Constraints: app size < 15–20 MB, offline-first, retry-safe, multi-language, assistive UX

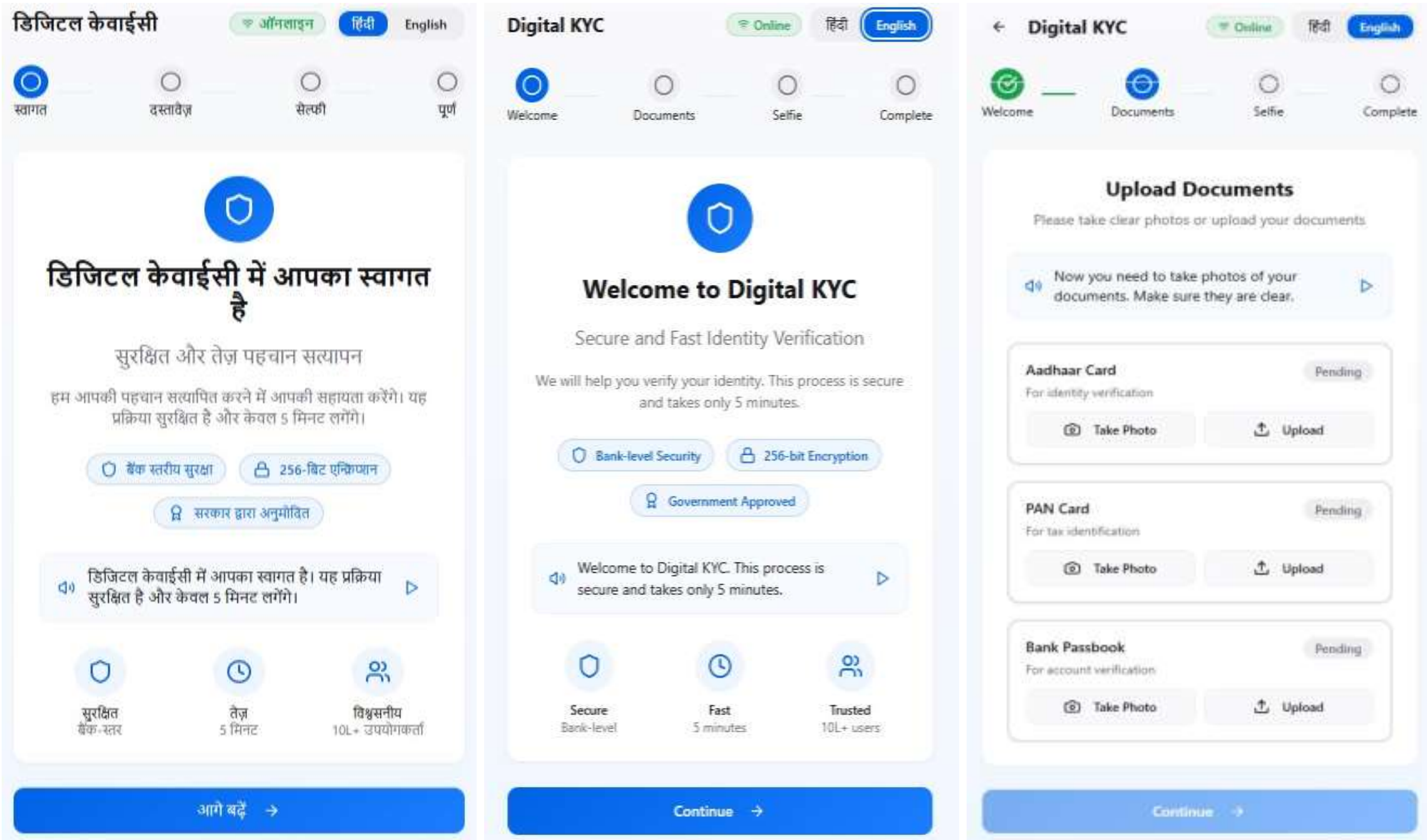
Design Principles

- Speak-their-language: 12+ Indian languages, voice prompts & large tap targets
- One-task-per-screen: progressive disclosure, clear success/failure states
- Never lose progress: offline queue + resumable uploads + background sync
- No dead-ends: help CTA, call-back option, and assisted mode for agents

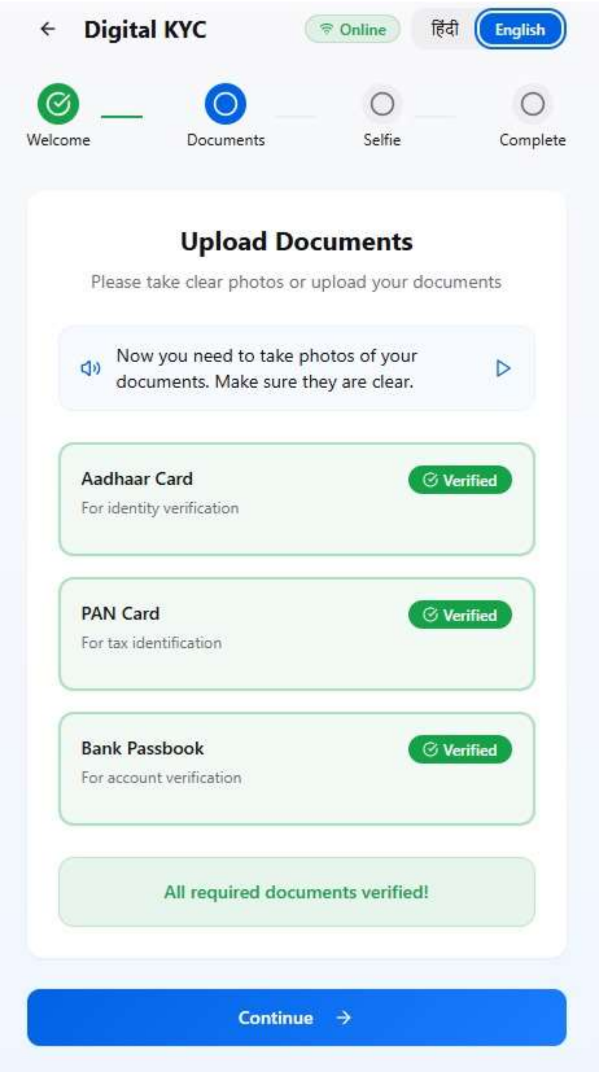
End-to-End User Flow (High Level)

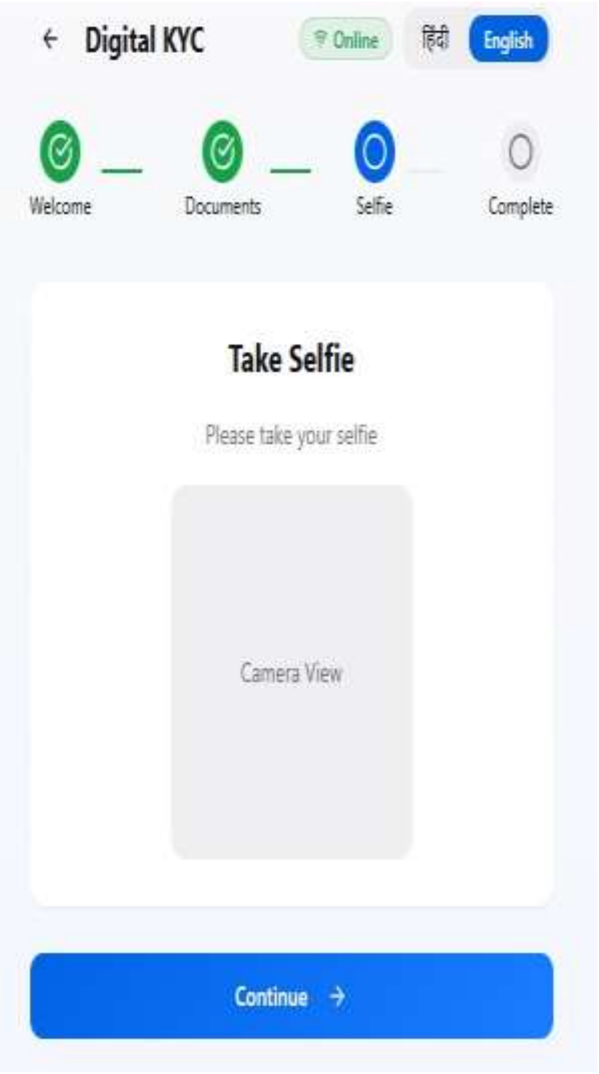
- 1) Language + consent → 2) Choose KYC path (Digilocker / Documents)
- 3) Face auth (liveness + face match) → 4) Review & confirm → 5) Success & share status
- Failures handled with: on-device guidance, retry timers, auto-resume, and assisted fallback

Wireframes: Home & Method Selection

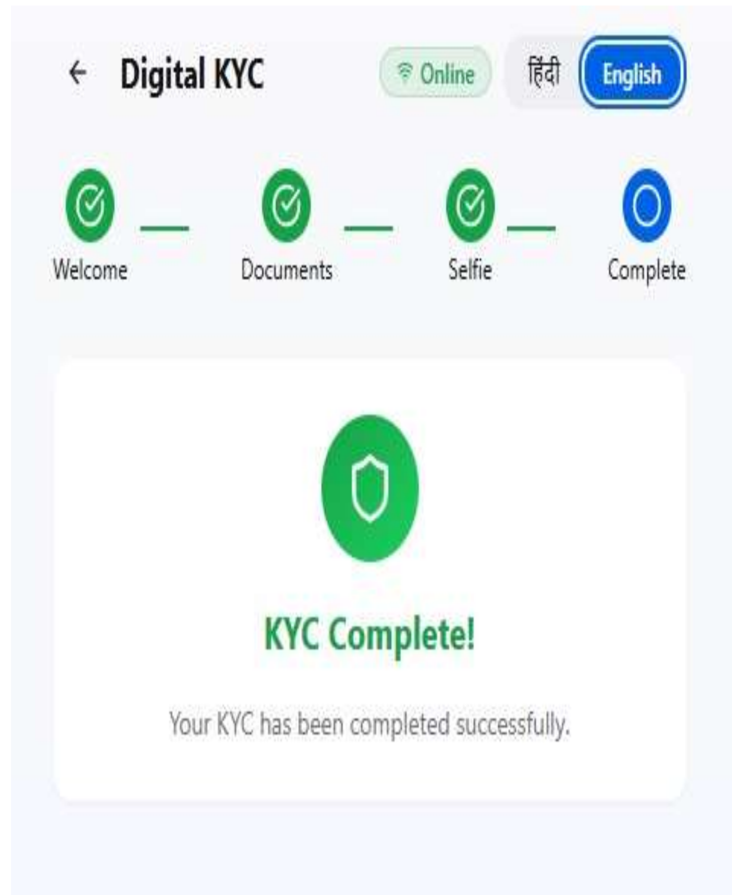


Wireframes: Document KYC Capture





Wireframes: Completion



Offline & Retry Design

- Local queue (encrypted) for steps + assets; show sync status per step
- Chunked uploads with checksum + HTTP range resume
- Backoff & scheduled retries; manual 'Try Again' with diagnostics
- Read-only fallback screens when server is unreachable

LLM-powered Innovations

- 1) Vernacular Copilot: voice-first guidance for each screen; detects confusion & offers help
- 2) Smart Error Coach: explains OCR/liveness failures in plain speech with exact next steps
- 3) Trust Layer: detect tampered documents textually (hallucination checks on extracted OCR)

Prioritisation Matrix

- High Impact / Low Effort: Offline queue + resumable uploads; simple Digilocker integration; large-tap UX
- High Impact / Medium Effort: On-device liveness; Smart Error Coach
- Medium Impact / Medium Effort: Vernacular Copilot; agent-assisted mode
- Low Impact / High Effort: Full on-device facematch; advanced document forgery detection

Technical Architecture (Overview)

- Mobile SDK: capture (docs/face), offline store, liveness, retry manager
- Edge workers: light OCR pre-checks; compression; PII redaction filters
- Backend: OCR/NLP, face match, liveness verify, Digilocker APIs, risk scoring, webhooks
- Integration: SDK (Android/iOS), Web-redirect, Partner callback URLs

On-device vs Server Trade-offs

- On-device: liveness prompts, capture guidance, compression → fast & offline-capable
- Server-side: facematch, final OCR/validation → accuracy, model updates, audit trail
- Hybrid switch based on bandwidth/CPU; fall back to server when device is weak

SDK Footprint Optimisation

- Modular downloads: feature flags (digilocker/doc/face) loaded on demand
- Shared native libs (NNAPI/MediaPipe), quantized models, WebP/AVIF assets
- Code shrink (R8/ProGuard), resource thinning, split APK/ABI

Bandwidth & Resume Strategy

- Image/video: HEVC/AV1 where supported; adaptive bitrate; delta re-uploads only
- Upload in chunks with integrity hashes; recover partial work after crashes
- State machine per user: idempotent APIs + step tokens for resume

Security & Compliance

- PII at rest: AES-256 on-device; passcode/biometric unlock; auto-wipe after sync
- In transit: TLS 1.3, cert pinning; signed requests with nonce + replay protection
- Least-privilege keys; device attestation; audit logs; consent receipts; data minimisation

Success Metrics & Experiments

- Primary: KYC completion rate, TTFK (time-to-finish KYC), drop-offs per step, cost per KYC
- Quality: false accept/reject, rework rate, selfie/doc retakes
- Reliability: offline completion %, avg retry count, p95 upload latency
- Run A/Bs: guidance variants, chunk sizes, liveness prompt sets

Rollout & Risks

- Pilot with agents; 3-language initial support; low-end device cohort first
- Risks: model drift, fraud patterns, broken network paths, privacy expectations
- Mitigation: shadow testing, kill switches, feature flags, rapid model rollback